Klimatkollen

Klimatkollen climate calculations 2025

Results

Klimatkollen has calculated its greenhouse gas emissions for 2025, according to the GHG Protocol (Greenhouse Gas Protocol). The table below shows the results for 2025, also compared with the emissions for 2024.

Scope, category and activities	kg CO2e 2024	kg CO2e 2025	Share 2025 (%)	Change 2024-2025	Relevance 2025
Scope 1 – Direct emissions	0	**0**			No activities in scope 1
Scope 2 – Indirect emissions from the production of purchased energy	0.0051	**0.0035**	**0.0001%**	-33%	Market-based chosen in summary
Market-based office electricity	0.0051	0.0035	0.0001%		Included
Location-based office electricity	0.6695	0.4500	0.0148%		Included
Scope 3 – Indirect emissions in the value chain	1511	**3047**	**100%**	102%	
3.1 Purchased goods and services	348	**1457**	**47.8%**	318%	Included
Server services	6.5	0.7	0.02%	-89%	
AI services	331	1402	46.0%	323%	
Chat completions	310	1359	44.6%	338%	
Embeddings	21	42.7	1.40%	103%	
Food and beverages	5.0	51.0	1.67%	920%	
Other purchased goods and services	5.5	3.2	0.11%	-42%	
3.2 Capital goods					No activities
3.3 Upstream fuel- and energy related activities	0.74	**0.50**	**0.02%**	-33%	Included, we use market-based
3.4 Upstream transportation and distribution					No activities
3.5 Waste generated in operations					Not calculated due to lack of data. Assumed to be negligibly small.
3.6 Business travel	1103	**1551**	**50.9%**	41%	Included
Train travel	2.9	118	3.9%	4041%	
Bus travel	0	14	0.5%		
Air travel	977	1296	42.5%		
Boat travel	123	**123**	**4.0%**	33%	
3.7 Employee commuting	39	**19**	**0.6%**	0%	Included
3.8 Upstream leased assets				-51%	Not relevant
3.9 Downstream transportation and distribution					Not relevant
3.10 Processing of sold products					Not relevant
3.11 Use of sold products	19.4	**19.4**	**0.6%**	0%	Included
3.12 End-of-life of sold products					Not relevant
3.13 Downstream leased assets					Not relevant
3.14 Franchises					Not relevant
3.15 Investments					Not relevant
Total emissions scope 1, 2 and 3	**1511**	**3047**		**102%**	

Analysis of the results

Klimatkollen has no activity in scope 1 and very small emissions in scope 2. All of Klimatkollen's climate impact is found in scope 3, where business travel (category 6) accounts for half (50.9%) of total emissions, of which five flights between Stockholm and Berlin account for 42.5% of the total, and boat and train travel each account for 4%.

The second-largest source of emissions (46%) comes from the use of AI services in the form of running OpenAI models, which are assumed to lead to electricity consumption in the USA. The remaining emissions (3.1%) come from meals during team days and board meetings, commuting to the office by bus, the server service at Berget AI, our office space, and use of our website (use of sold products).

During 2025, data was collected for the same categories as in 2024, making the figures comparable between the years. Between 2024 and 2025, Klimatkollen's emissions increased by 102% (from 1511 kWh to 3047 kWh). This is mainly due to more AI services, more business trips made by air, more train journeys on the continent (with operators that have higher emissions than SJ), and more team days with meals.

Methodology description

For the calculation of Klimatkollen's 2025 climate accounts, available input data was collected and matched with a suitable emission factor. See the table below for details per activity.

Scope and category	Type of data	Data quality	Source emission factor	Comment
Scope 2				
Market-based	Electricity consumption (kWh)	Measured	GodEl (2024)	Electricity usage data from the landlord. Estimated based on the number of offices spaces used by Klimatkollen in their office building.
Location-based	Electricity consumption (kWh)	Measured	AIB (2024)	Electricity usage data from the landlord. Estimated based on the number of offices spaces used by Klimatkollen in their office building.
Scope 3				
3.1 Purchased goods and services	Number of purchased business cards and reports (units), energy consumption of server services (kWh), energy consumption of AI-usage (kWh)	Estimated/ Measured	GodEl (2024)/AIB (2018), WWF (2025), SIK (2021), Idemat (2025), CaDI (2025)	Includes all purchases of goods and services. Energy consumption comprises server energy provided by the supplier Berget as well as energy use from AI services (OpenAI models; see detailed methodology below). The weight (kg) of purchased business cards and reports is estimated based on quantities and an assumed weight per unit.
3.3 Fuel- and energy related activities	See Scope 2	Measured	See Scope 2	
3.5 Waste generated in operations	Not available	-	-	Data on the weight or cost of generated waste was not available from the property owner in 2025.
3.6 Business travel	Travelled distance (km & nautical mile)	Measured	SJ (2025), NTM (2021), Destination Gotland (2025)	
3.7 Employee commuting	Commuting distance (km)	Measured	Energimyndigheten (2023), Klimatsmart semester (2024)	
3.11 Use of sold products	Time on site (seconds)	Measured		The use of the Klimatkollen website is calculated and treated as a service, although it is not a sold service. This activity is nevertheless included, as it is considered essential to Klimatkollen’s core purpose of providing climate data to the public.

**Scope 2 has been calculated using both the market-based and location-based methods. The market-based method was used for the summary of results.*

Calculation of AI services

The method used to calculate the carbon dioxide emissions linked to Klimatkollen's use of AI services (OpenAI) is based on a scientific article** written by Francisco Caravaca et al. In the article, the authors identify, among other things, linear relationships between energy consumption and the number of input and output tokens in a prompt. The results show that energy consumption increases by a factor of 2.19 when the number of input tokens increases from 100 to 900, and by a factor of 11 when the number of output tokens increases within the same interval. In addition, the authors present the energy consumption for a so-called "typical prompt" for a range of different models.

Klimatkollen's data consists of daily reports on the number of model runs performed and the total number of input and output tokens. For each reported day, an average input and output token count per prompt has been calculated and scaled up based on the number of runs per day. Based on this, the energy consumption per prompt has been estimated using the linear relationships and the energy consumption values presented in Caravaca et al. The exact models used by Klimatkollen are not included in the article; instead, two proxy models of similar size to Klimatkollen's have been selected: a smaller model (30 billion parameters) and a larger model (200 billion parameters).

Two different types of model runs are evaluated: "chat completions," which correspond to typical text-based interactions between people and models, and "embeddings," which convert input texts into mathematical vector-based representations. Embeddings are less energy-intensive than chat completions, by a factor of ten.

There are uncertainties regarding the geographical location of the OpenAI data centers used for the relevant model runs. To apply a more conservative assumption, all electricity use related to Klimatkollen's use of AI services has been assumed to take place in data centers located in the USA. The calculation only covers the direct electricity use for the model runs performed and therefore does not include emissions from training AI models, the operation and maintenance of data centers, or embodied emissions from hardware such as GPUs.

**https://arxiv.org/pdf/2511.05597